
Hidden-Feature Matching and Movement Estimates Diverge in a Gait JEPA

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Abstract

1 We evaluate whether learning to predict hidden skeleton features improves access
2 to a movement quantity in human gait. A small skeleton Joint-Embedding Predic-
3 tive Architecture (JEPA) is trained without movement labels, then frozen while a
4 separate regression estimates a signed left–right speed contrast. In source-video
5 cross-validation on 625 clips from 93 videos, trained predictors more consistently
6 favor hidden features from the correct clip over another video’s features. With
7 the same expanded summary and regression procedure, however, mean source-
8 balanced R^2 falls from 0.22 at initialization to 0.10–0.11 after training. Motion
9 and region masks leave their advantage over matched random masks uncertain.
10 This focused evaluation shows why the success of latent prediction should be as-
11 sessed alongside the usefulness of features for a specified observable of an evol-
12 ving system. The measured gap concerns whole-clip gait representations; future
13 movement prediction remains untested on real data.

14 1 Introduction

15 Walking is an evolving system: coupled body landmarks change position over time, and movement
16 depends on their trajectories. Gait rhythm and coordination motivate temporal modeling, although
17 video-derived coordinates describe only part of the physical process. Temporal world modeling
18 aims to represent such a system well enough to predict its observable behavior. We test access to
19 one movement measurement within that larger objective.

20 We study that question with S-JEPA [1], using a left–right contrast of median coordinate speeds.
21 Clinical gait asymmetry motivates the setting [2]; our contrast has no clinical validation. Our central
22 null is no improvement from JEPA training over matched initial weights for this readout on held-
23 out source videos. The no-improvement null also motivated the registered reflection study. The
24 expanded readout and mask questions developed on the same cohort and remain exploratory.

25 The contribution is a domain-specific evaluation relevant to temporal foundation models: success on
26 latent prediction needs to be considered alongside access to observable movement. Scaling, broad
27 transfer, persistent temporal state and gait simulation remain outside the demonstrated scope.

28 2 Study design

29 **Data and observable.** Quality checks retain 625 GAVD clips from 93 source videos [3]. Each
30 archive has shape $T_i \times 33 \times 4$: three estimated coordinates plus landmark visibility, with original
31 frame numbers and frame rate. Coordinate units are image normalized with inferred depth; they are
32 not calibrated physical speeds.

33 The movement path uses observed coordinates, pelvis centering and a body-width scale, without
 34 gap filling or resizing. For five pairs—shoulders, knees, ankles, heels and foot tips—we calculate
 35 median speeds $m_{L,k}, m_{R,k}$ using original timestamp differences. Both sides must be observed at
 36 both ends of at least eight common transitions for every pair. The signed target is

$$y = \frac{1}{5} \sum_{k=1}^5 \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + 10^{-8}}.$$

37 Anatomical reflection exchanges left/right landmarks and validity flags and flips the centered hor-
 38 izontal coordinate, giving $y(Mx) = -y(x)$. Hips define the pelvis and are excluded from the
 39 contrast because centering equalizes their speed magnitudes. The statistic describes relative median
 40 speed, without measuring phase, stride timing or loading. Reversing a trajectory with its sampling
 41 intervals preserves speed magnitudes, so recovering this target alone cannot demonstrate learned
 42 temporal dynamics.

43 **Encoder input.** A landmark is observed when visibility is at least 0.45 and all coordinates are finite.
 44 A separate encoder path fills interior gaps of at most four samples, treating filled positions as valid.
 45 After pelvis centering and body-width normalization, coordinates and validity are linearly resized to
 46 64 positions by sequence index. Original time intervals are discarded. Resized validity must reach
 47 0.999; invalid coordinates are zero. Each joint’s four-position patch projects 12 coordinates to 96
 48 features, giving a 16×33 grid. A patch is valid only if all four positions are valid. Anatomical
 49 identities and prepared positions survive this operation; the original trajectory, duration and cadence
 50 need not.

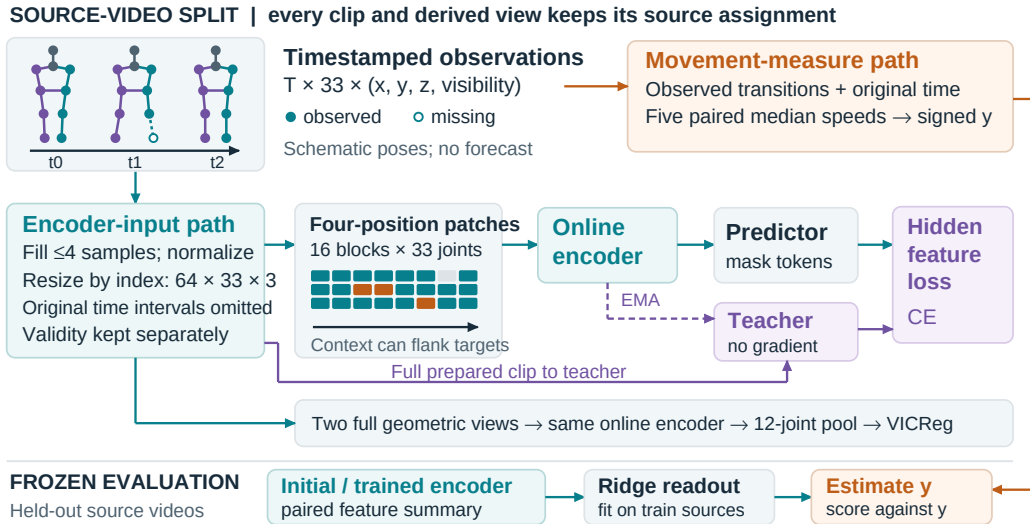


Figure 1: Two measurement paths and the completed JEPA training grid. Poses and mask cells are schematic. The movement target reaches only the separate regression evaluation. The online encoder has visible context on either side of a hidden position; the teacher and full-view regularizer access complete prepared clips.

51 **Training.** The grid uses 1,200 AdamW updates per encoder (architecture and settings in Appendix
 52 A). The online encoder zeroes hidden patch content before adding joint and position embeddings;
 53 the predictor substitutes learned mask tokens at target positions. Cross-entropy matches centered
 54 teacher feature distributions at hidden valid positions, averaged within clips and then across clips.
 55 A teacher follows the online weights by exponential moving average without gradients. A full-view
 56 VICReg term [4], weighted 0.05, pools twelve gait joints across two geometric views. Thus the total
 57 objective also accesses complete clips. Training uses CUDA BF16 and evaluation FP32.

58 We compare three motion conditions and two region conditions across five folds and five seeds: 125
 59 encoders. The motion conditions use random sampling, a MAMP-style sampler [5], and a robust

60 motion mixture. Motion masks sample across all 33 joints using a count derived from the twelve-
 61 joint gait pool. Connected regions hide six connected landmarks over eight of sixteen blocks, with a
 62 separate count-matched random reference. Each comparison matches initial weights, source draws,
 63 views and update counts. No reflection augmentation or reflection loss appears in this grid.

64 **Held-out readout.** Five outer folds keep each video’s clips and derived views together. Fresh
 65 encoders fit only outer-training videos with source-balanced sampling. All overlapping windows,
 66 mirrors and augmentations inherit the source split. The encoder is frozen while ridge regression, a
 67 linear model with a coefficient penalty, estimates y . Three inner source folds select the penalty using
 68 only outer-training labels; imputation and scaling are fitted within each inner training set and then
 69 refitted with the regression on all outer-training sources. Inner-validation clips may have contributed
 70 without labels to outer-training encoder fitting.

71 Both encoders use bilateral mean features (960 inputs), expanded with standard deviations, absolute
 72 adjacent-block feature changes and ten validity-support fractions (2,890 inputs). We pool outer-
 73 test predictions within each seed, weighting each video equally, then average the five R^2 values.
 74 Paired 95% source-bootstrap intervals resample videos with their clips and seed predictions together
 75 (2,000 resamples). They condition on the fitted models and existing splits, excluding uncertainty
 76 from retraining, alternative partitions and development choices. Repeated seeds and mask draws do
 77 not add independent videos.

78 3 Findings

79 **Movement readout falls after training.** The expanded initial representation reaches mean $R^2 =$
 80 0.223, compared with 0.101–0.114 for trained teachers (Figure 2). All five conditions decrease in
 81 every seed, with greater mean absolute error. Paired trained-minus-initial intervals are below zero
 82 in every condition. Final online encoders also score below their matched initial encoders.

	Training mask	Mean R^2	ΔR^2	95% source interval
	Random (motion)	0.114	-0.109	[-0.170, -0.041]
	MAMP-style	0.113	-0.110	[-0.165, -0.051]
	Motion mixture	0.114	-0.108	[-0.171, -0.042]
	Random (region)	0.109	-0.113	[-0.166, -0.060]
	Connected region	0.101	-0.122	[-0.183, -0.055]

84 Table 1: trained teacher minus matched initialization. Intervals resample source videos, conditional
 85 on fitted models; differences are calculated before rounding.

86 Expanding the initial readout raises R^2 from 0.071 to 0.223 (+0.152). The added components mix
 87 temporal variation, absolute increments and observation support; standard deviation discards order,
 88 and absolute increments discard direction. Their combined gain therefore does not isolate sensi-
 89 tivity to temporal order. The initial control retains the pretrained pose detector, input preparation,
 90 anatomical identities and supervised regression.

91 **Correct-clip features become easier to match.** Each diagnostic compares source-weighted mean
 92 squared feature error against the correct clip and valid targets from another source, holding the eval-
 93 uated clips and hidden positions fixed. All 375 trained checks favor correct targets; 33 of 75 initial
 94 checks do so. The trained checks cover 125 fitted models under three evaluation masks (left-leg gap,
 95 right-leg gap and scattered gap). Initial checks cover 25 matched fold/seed controls under the same
 96 three masks, with copies shared between experiment families counted once. Visible context and
 97 full-clip teacher targets share pose, viewpoint and missingness, as well as motion. Those cues can
 98 support correct-clip matching without isolating hidden movement. Teachers and target distributions
 99 change during training, so raw training losses cannot rank representation quality across conditions.

100 **Changing mask geometry does not establish better readout.** The inspection draws confirm that
 101 motion samplers select higher-motion targets and that connected regions remove immediate tempo-
 102 ral brackets. Motion masks hide approximately 17% of valid all-landmark tokens; regions hide 9.9%.
 103 Their readout comparisons therefore use distinct random references. All three mask-comparison
 104 intervals span zero (Appendix B). Their benefits remain uncertain; the intervals do not establish
 105 equivalence.

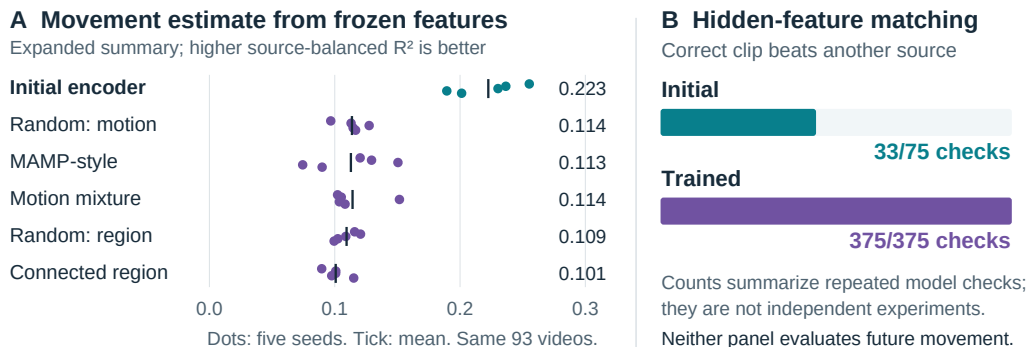


Figure 2: Different outcomes from the same fitted models. A: frozen movement readout; points are five seed scores, and short vertical marks their mean. B: correct-clip feature error is lower in 375 of 375 trained diagnostic checks, versus 33 of 75 distinct initial checks. These checks reuse models and videos.

Preparing input changes the measurement. A saved comparison recalculates the target after input preparation: 623 clips from 92 sources have finite values on both paths, with source-weighted sign agreement 70.4% and calculation-agreement $R^2 = 0.218$. This is agreement between two calculations of the observable, not a learned model score. Neither the achievable readout accuracy nor an individual preprocessing cause follows from it.

4 Interpretation and next experiment

The predictor favors the correct clip more consistently after training, while the same regression procedure estimates its left-right speed contrast less accurately from trained features. Poorer readout performance does not imply complete destruction of movement information. In particular, 49/125 teacher fits and 96/125 online fits select the largest ridge penalty, 10,000. Wider training-only selection, support-only controls and summaries of matched dimension would help separate feature geometry from readout limitations. Source holdout does not establish participant independence, chronological holdout or an untouched development cohort. Recording and pose-estimation cues remain possible explanations.

A next experiment would decode future coordinates from past-only inputs at common valid end-points and physical-time horizons. Fit and select decoders on training sources only; compare last-position, constant-velocity, direct-pose, initial-encoder and mismatched-training-future controls. Future coordinates and visibility must not change prefix preparation, normalization, masks or predictions. The retained past-feature readout and future-feature decoding studies have synthetic training evidence; the real-data horizon census establishes eligibility only. Decoding observed future-teacher features diagnoses the decoder; decoding features predicted from the past evaluates forecasts. Multiple horizons from one prefix are not recursive rollouts.

The finding is conditional on this cohort, recipe and readout. Forecasting and simulation require tests of future behavior beyond the whole-clip contrast.

5 Related work

Predictive representation learning selects which variation to retain through both its target and its context. Skeleton methods predict learned features [1] or explicit motion [5]; connected-body masks reduce nearby cues [6]. Action recognition evaluates the resulting features for labels, while video anticipation and action-conditioned planning assess other behavior [7]. Work separating invariant and transformation-sensitive features further exposes this dependence on the task [8]. Our gait contrast adds an observable-specific test with matched initial weights. The mask variants borrow sampling ideas, without experimentally comparing those complete methods or their reported capabilities.

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171 A Measurement and implementation details

172 Of 666 selected annotations, 642 match pose archives and 625 pass quality checks. The motion
173 target count is half the batch’s smallest valid twelve-joint gait pool, rounded down, then sampled
174 across all 33 joints.

175 The completed recipe uses 96 features, encoder/predictor depths 4/2, four attention heads, batch size
176 20, 1,200 AdamW updates, learning rate 0.001, weight decay 0.05 and teacher momentum 0.999.

177 The archive coordinates are $x = (x_0 + uW_c)/W$, $y = (y_0 + vH_c)/H$, and $z = dW_c/W$, where crop
178 origin and size are (x_0, y_0) , W_c , H_c , frame size is W , H , and u, v, d are pose estimates. Unequal
179 image-axis scaling and inferred depth remain after body normalization. The scale is the median of
180 valid shoulder and hip widths. The movement path requires observed hips; the encoder uses the
181 median valid pelvis midpoint for missing hips (zero if none is available). No boundary extrapolation
182 fills gaps.

183 The teacher receives the unaugmented prepared clip; the masked online path receives a geometric
184 view. Two full online views use rotations up to 8 degrees in the horizontal-depth plane and transla-
185 tions up to 0.03 in image-plane coordinates. Target/predictor temperatures are 0.06/0.10; the teacher
186 center uses momentum 0.9. Gradient norm is clipped at 1. The twelve-joint regularizer pool is
187 shoulders, hips, knees, ankles, heels and foot tips. Its projector is a separate training component.
188 It combines invariance, variance and covariance penalties with internal weights 25, 25 and 1, then
189 overall weight 0.05.

190 The expanded summary uses common left/right validity within each pair; increments require con-
191 secutive common valid blocks. Each pair contributes differences and sums of means, standard de-
192 viations and mean absolute feature increments (six 96-vectors), plus two support fractions. Thus
193 $5 \times 6 \times 96 + 10 = 2,890$. Missing support gives zero summary values. Temporal positions in
194 this summary are prepared block indices, not original timestamps. Ridge penalties are selected from
195 0.01, 0.1, 1, 10, 100, 1000, 10000.

196 Source-balanced R^2 uses weights $w_{vi} = 1/(Vn_v)$ and weighted test mean $\bar{y}_w$:

$$R_w^2 = 1 - \frac{\sum_{vi} w_{vi} (y_{vi} - \hat{y}_{vi})^2}{\sum_{vi} w_{vi} (y_{vi} - \bar{y}_w)^2}.$$

197 The test mean defines this metric’s denominator, not a deployable fitted baseline. Outer test folds
198 contain 19, 19, 19, 18 and 18 videos and 189, 182, 72, 77 and 105 clips. The retained provenance
199 audit checked archive IDs and hashes, source membership and absence of test-source training draws.
200 Those checks concern file integrity and split adherence, not cross-upload content deduplication. Per-
201 sistent person IDs and complete annotation retrieval/version records are unavailable, preventing a
202 person-level contamination audit or full reconstruction of acquisition history. Source holdout alone
203 provides no evidence of generalization under distribution shift.

204 B Supporting results and reproducibility

205 MAMP-style, motion-mixture and region masks minus their own random references give
206 $\Delta R^2 = -0.0008, +0.0005$ and -0.009 , with intervals $[-0.020, 0.015]$, $[-0.015, 0.015]$ and
207 $[-0.033, 0.018]$. A 264-value coordinate/validity summary reaches $R^2 = 0.035$; it is a limited
208 direct-pose baseline, not a ceiling on coordinate information.

209 These exploratory percentile intervals use bootstrap seed 812, retain all five fitted seeds and have no
210 familywise or development-selection adjustment. The numerical supplement preserves aggregate
211 seed scores and recorded intervals. The executed report identifies 125,000 prediction rows and 200
212 pooled score rows; a saved audit records their recomputation. Raw predictions and checkpoints
213 are unavailable in the revision environment, so this revision checks aggregate consistency without
214 claiming to repeat that bootstrap or inference.

215 A separate reflection-augmentation comparison has real-data summary evidence: augmentation re-
216 duces the specified teacher-token reflection discrepancy by $\Delta q = -0.008 [-0.010, -0.007]$, while
217 its readout difference is $\Delta R^2 = 0.004 [-0.006, 0.013]$. This uses its own recipe, four inner readout
218 folds and compact summary; it is not pooled with the motion grid. An explicit reflection loss has

219 synthetic evidence only. Likewise, a readout forced to take the form $g(x) = [h(x) - h(Mx)]/2$
220 reverses sign algebraically without establishing learned movement information.

221 GAVD annotations describe the cohort, not affected side or treatment response. Uncalibrated coordi-
222 nates and absent clinical reference measurements preclude a clinical interpretation of y . Public
223 annotation access does not authorize redistribution of separately hosted videos or derived poses;
224 neither is included in the supplement.

225 **AI assistance.** An AI assistant supported author-side evidence reconciliation, literature checking,
226 manuscript revision, figure programming and adversarial checks. No new model training was per-
227 formed for this revision. Responsibility for the submitted text, figures, claims and references remains
228 with the authors.